

20



► More generally for non-linear paths:

$$\lambda(\beta) = \frac{1}{2} \mathbb{E}[| \dot{\pi}_{\beta}(X') - \dot{\pi}_{\beta}(X) |], \quad (X, X') \sim \pi_{\beta} \otimes \pi_{\beta}$$

$$\lambda(\beta) = \frac{1}{2} \mathbb{E}[|V(X') - V(X)|], \quad (X, X') \sim \pi_\beta \otimes \pi_\beta$$

► **Theorem:** If V has third moments with respect to π_0 and π_1 , then $\beta \mapsto \lambda(\beta)$ is a twice-continuously differentiable function and satisfies:

► Moreover for all $\beta, \beta' \in [0, 1]$ with $\Delta\beta = \beta' - \beta$

$$\left| r(\beta, \beta') - \int_{\beta}^{\beta'} \lambda(\tilde{\beta}) d\tilde{\beta} \right| \leq C |\Delta\beta|^3 \sup_{\tilde{\beta} \in [\beta, \beta']} \pi_{\tilde{\beta}}[|V|^3]$$

► The **local barrier** $\lambda(\beta)$ at β as the instantaneous rate of rejection:

$$\lambda(\beta) = \lim_{\Delta\beta \rightarrow 0} \frac{r(\beta, \beta + \Delta\beta)}{|\Delta\beta|}$$

► Measures how sensitive the annealing path is to small perturbations at β

► Defines the speed of the path with respect to the rejection!

LOCAL BARRIER

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GLOBAL BARRIER